

| Features                                                                                                                                                                                                                                                                                                                                                        | Model                                                                     | K-fold MAE         | K-fold MSE     | K-fold RMSE | K-fold R2 | Y-test MAE | Y-test MSE    | Y-test RMSE | Y-test R2 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|--------------------|----------------|-------------|-----------|------------|---------------|-------------|-----------|
| No AI, w/ WorkExp and JobSat                                                                                                                                                                                                                                                                                                                                    | LinearRegression()                                                        | 20568.543          | 780428037.998  | 27936.142   | 0.4673    | 17649.148  | 580915498.879 | 24102.189   | 0.5144    |
| ['WorkExp', 'JobSat', 'LearnCodeChoose', 'PurchaseInfluence', 'DatabaseChoice', 'PlatformChoice', 'WebframeChoice', 'DevEnvsChoice', 'Country', 'EdLevel', 'DevType', 'OrgSize', 'ICorPM', 'RemoteWork', 'Industry', 'LanguageChoice', 'lang_group_web_frontend', 'lang_group_data_science', 'lang_group_query_shell', 'lang_group_jvm', 'lang_group_dotnet']   | DecisionTreeRegressor (max_depth=5, min_samples_leaf=5, random_state=42)  | 22095.447          | 923214331.061  | 30384.442   | 0.3648    | 19945.142  | 674469258.903 | 25970.546   | 0.4361    |
|                                                                                                                                                                                                                                                                                                                                                                 | DecisionTreeRegressor (max_depth=6, min_samples_leaf=10, random_state=42) | 22159.146          | 915535823.658  | 30257.823   | 0.3824    | 20510.664  | 714025744.576 | 26721.260   | 0.4031    |
|                                                                                                                                                                                                                                                                                                                                                                 | Lasso(alpha=0.01)                                                         | 21356.686          | 855166229.621  | 29243.225   | 0.4270    | 19322.806  | 663034235.707 | 25749.451   | 0.4457    |
|                                                                                                                                                                                                                                                                                                                                                                 | ElasticNet(alpha=0.01, l1_ratio=0.5)                                      | 20603.413          | 800044145.757  | 28285.052   | 0.4662    | 18449.144  | 624576077.467 | 24991.520   | 0.4779    |
|                                                                                                                                                                                                                                                                                                                                                                 | Ridge(alpha=1.0)                                                          | 20466.111          | 772402416.450  | 27792.129   | 0.4741    | 17608.104  | 579920560.826 | 24081.540   | 0.5152    |
|                                                                                                                                                                                                                                                                                                                                                                 | HuberRegressor (epsilon=1.35, max_iter=1000)                              | 20744.775          | 787328378.440  | 28059.372   | 0.4598    | 17807.791  | 595073762.434 | 24394.134   | 0.5025    |
|                                                                                                                                                                                                                                                                                                                                                                 | LinearRegression()                                                        | 22436.159          | 899651894.274  | 29994.198   | 0.3692    | 18053.130  | 586990313.702 | 24227.883   | 0.5093    |
| [ 'LearnCodeChoose', 'PurchaseInfluence', 'DatabaseChoice', 'PlatformChoice', 'WebframeChoice', 'DevEnvsChoice', 'Country', 'EdLevel', 'DevType', 'OrgSize', 'ICorPM', 'RemoteWork', 'Industry', 'LanguageChoice', 'lang_group_web_frontend', 'lang_group_data_science', 'lang_group_query_shell', 'lang_group_jvm', 'lang_group_dotnet', 'lang_group_systems'] | DecisionTreeRegressor (max_depth=5, min_samples_leaf=5, random_state=42)  | 24646.416          | 1103368043.582 | 33216.984   | 0.2256    | 22206.035  | 846077375.499 | 29087.409   | 0.2927    |
|                                                                                                                                                                                                                                                                                                                                                                 | DecisionTreeRegressor (max_depth=6, min_samples_leaf=10, random_state=42) | 24430.588          | 1054146172.279 | 32467.617   | 0.2538    | 21385.931  | 769869672.405 | 27746.525   | 0.3564    |
|                                                                                                                                                                                                                                                                                                                                                                 | Lasso(alpha=0.01)                                                         | 23167.379          | 975134003.080  | 31227.136   | 0.3260    | 20163.782  | 703686879.114 | 26527.097   | 0.4117    |
|                                                                                                                                                                                                                                                                                                                                                                 | ElasticNet(alpha=0.01, l1_ratio=0.5)                                      | 22287.670          | 901732010.835  | 30028.853   | 0.3738    | 19088.160  | 640708624.144 | 25312.223   | 0.4644    |
|                                                                                                                                                                                                                                                                                                                                                                 | Ridge(alpha=1.0)                                                          | 22323.874          | 891221192.142  | 29853.328   | 0.3751    | 18032.938  | 585088912.539 | 24188.611   | 0.5109    |
|                                                                                                                                                                                                                                                                                                                                                                 | HuberRegressor (epsilon=1.35, max_iter=1000)                              | 22677.946          | 915672306.498  | 30260.078   | 0.3555    | 18380.796  | 608179489.620 | 24661.295   | 0.4916    |
|                                                                                                                                                                                                                                                                                                                                                                 | LinearRegression()                                                        | 22779.342          | 1028937901.759 | 32077.062   | 0.3408    | 19630.759  | 952030080.651 | 30854.985   | 0.2041    |
| With AI, WorkExp and JobSat                                                                                                                                                                                                                                                                                                                                     | DecisionTreeRegressor (max_depth=5, min_samples_leaf=5, random_state=42)  | 22348.738          | 939977932.832  | 30659.060   | 0.3577    | 20014.191  | 677772033.607 | 26034.055   | 0.4334    |
|                                                                                                                                                                                                                                                                                                                                                                 | DecisionTreeRegressor (max_depth=6, min_samples_leaf=10, random_state=42) | 22405.664          | 929937911.922  | 30494.883   | 0.3744    | 20913.351  | 736869773.011 | 27145.345   | 0.3840    |
|                                                                                                                                                                                                                                                                                                                                                                 | Lasso(alpha=0.01)                                                         | 21226.117          | 847924277.586  | 29119.139   | 0.4321    | 19354.662  | 662459422.547 | 25738.287   | 0.4462    |
|                                                                                                                                                                                                                                                                                                                                                                 | ElasticNet(alpha=0.01, l1_ratio=0.5)                                      | 20465.653          | 794187919.320  | 28181.340   | 0.4701    | 18560.616  | 623237857.099 | 24964.732   | 0.4790    |
|                                                                                                                                                                                                                                                                                                                                                                 | Ridge(alpha=1.0)                                                          | 21344.379          | 831799941.813  | 28840.942   | 0.4368    | 18398.520  | 663605631.985 | 25760.544   | 0.4452    |
|                                                                                                                                                                                                                                                                                                                                                                 | HuberRegressor (epsilon=1.35, max_iter=1000)                              | failed to converge |                |             |           |            |               |             |           |
|                                                                                                                                                                                                                                                                                                                                                                 | LinearRegression()                                                        | 22779.342          | 1028937901.759 | 32077.062   | 0.3408    | 19630.759  | 952030080.651 | 30854.985   | 0.2041    |

| Features                                                                                                                                                                                                                                                                                                                                                                                                                | Model                                                                              | K-fold MAE         | K-fold MSE         | K-fold RMSE | K-fold R2 | Y-test MAE | Y-test MSE     | Y-test RMSE | Y-test R2 |  |  |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------|--------------------|-------------|-----------|------------|----------------|-------------|-----------|--|--|--|
| 'AIThreat', 'AIComplex',<br>'AIAgentKnowledge',<br>'AIAgentOrchestration',<br>'AIAgentChoice', 'Country',<br>'EdLevel', 'DevType', 'OrgSize',<br>'ICorPM', 'RemoteWork',<br>'Industry',<br>'LanguageChoice',<br>'AIThreat', 'AIComplex',<br>'AIAgentKnowledge'                                                                                                                                                          | HuberRegressor<br>(epsilon=1.35,<br>max_iter=5000)                                 | 22917.633          | 1011418344.535     | 31802.804   | 0.330     | 20085.386  | 1134258963.446 | 33678.761   | 0.052     |  |  |  |
| With AI, No WorkExp, No JobSat                                                                                                                                                                                                                                                                                                                                                                                          | LinearRegression()                                                                 | 24946.567          | 1235091085.093     | 35143.863   | 0.2344    | 20926.457  | 1095972523.768 | 33105.476   | 0.0838    |  |  |  |
| ['LearnCodeAI',<br>'LearnCodeChoose',<br>'PurchaseInfluence',<br>'DatabaseChoice',<br>'PlatformChoice',<br>'WebframeChoice',<br>'DevEnvsChoice',<br>'AIModelsChoice',<br>'AISelect', 'AIAgents',<br>'Country', 'EdLevel',<br>'DevType', 'OrgSize',<br>'ICorPM', 'RemoteWork',<br>'Industry',<br>'LanguageChoice',<br>'AIThreat', 'AIComplex',<br>'AIAgentKnowledge']                                                    | DecisionTreeRegressor<br>(max_depth=5,<br>min_samples_leaf=5,<br>random_state=42)  | 24798.750          | 1116959475.621     | 33420.944   | 0.2197    | 21713.381  | 811344696.178  | 28484.113   | 0.3217    |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                         | DecisionTreeRegressor<br>(max_depth=6,<br>min_samples_leaf=10,<br>random_state=42) | 24452.378          | 1060200706.304     | 32560.723   | 0.2505    | 21279.302  | 755729531.884  | 27490.535   | 0.3682    |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                         | Lasso(alpha=0.01)                                                                  | 23114.008          | 970306206.589      | 31149.738   | 0.3306    | 20147.473  | 704551284.495  | 26543.385   | 0.4110    |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                         | ElasticNet(alpha=0.01,<br>l1_ratio=0.5)                                            | 22217.539          | 895764289.664      | 29929.322   | 0.3803    | 19094.971  | 643069806.129  | 25358.821   | 0.4624    |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                         | Ridge(alpha=1.0)                                                                   | 23349.006          | 966194780.505      | 31083.674   | 0.3382    | 19360.759  | 771285577.697  | 27772.029   | 0.3552    |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                         | HuberRegressor<br>(epsilon=1.35,<br>max_iter=1000)                                 | 25209.744          | 1234665626.190     | 35137.809   | 0.2118    | 22007.281  | 1331290369.499 | 36486.852   | -0.1130   |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                         | LinearRegression()                                                                 | 151582.453         | 29040434913115.400 | 5388917.787 | 0.2137    | 17913.911  | 599391991.874  | 24482.483   | 0.4989    |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                         | DecisionTreeRegressor<br>(max_depth=5,<br>min_samples_leaf=5,<br>random_state=42)  | 22562.374          | 954355183.051      | 30892.640   | 0.3591    | 19836.702  | 682836909.272  | 26131.148   | 0.4291    |  |  |  |
| ['WorkExp', 'YearsCode',<br>'ToolCountWork',<br>'ToolCountPersonal',<br>'LearnCodeChoose',<br>'PurchaseInfluence',<br>'LanguageChoice',<br>'DatabaseChoice',<br>'PlatformChoice',<br>'WebframeChoice',<br>'DevEnvsChoice',<br>'AIModelsChoice',<br>'AISelect', 'AIAgents',<br>'Country', 'EdLevel',<br>'DevType', 'OrgSize',<br>'ICorPM', 'RemoteWork',<br>'Industry', 'Employment',<br>'LanguageHaveWorkedWith_count'] | DecisionTreeRegressor<br>(max_depth=6,<br>min_samples_leaf=10,<br>random_state=42) | 22995.066          | 978365388.106      | 31278.833   | 0.3458    | 19956.211  | 684062984.612  | 26154.598   | 0.4281    |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                         | Lasso(alpha=0.01)                                                                  | 167250.032         | 35835463732033.8   | 5986272.942 | 0.1450    | 19995.824  | 693354808.591  | 26331.631   | 0.4204    |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                         | ElasticNet(alpha=0.01,<br>l1_ratio=0.5)                                            | 236504.528         | 78671983445781.500 | 8869722.851 | 0.1586    | 18947.593  | 640699972.061  | 25312.052   | 0.4644    |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                         | Ridge(alpha=1.0)                                                                   | 154290.201         | 30300490644553.000 | 5504588.145 | 0.2178    | 17861.087  | 595455080.978  | 24401.948   | 0.5022    |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                         | HuberRegressor<br>(epsilon=1.35,<br>max_iter=1000)                                 | failed to converge |                    |             |           |            |                |             |           |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                         | HuberRegressor<br>(epsilon=1.35,<br>max_iter=5000)                                 | 30995.686          | 178441354326.218   | 422423.193  | 0.332     | 17977.860  | 606965945.999  | 24636.679   | 0.493     |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                         | LinearRegression()                                                                 | 151582.453         | 29040434913115.400 | 5388917.787 | 0.2137    | 17913.911  | 599391991.874  | 24482.483   | 0.4989    |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                         | DecisionTreeRegressor<br>(max_depth=5,<br>min_samples_leaf=5,<br>random_state=42)  | 22562.374          | 954355183.051      | 30892.640   | 0.3591    | 19836.702  | 682836909.272  | 26131.148   | 0.4291    |  |  |  |